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AMRITA VISHWA VIDYAPEETHAM

AHEAD Online

Cryptography

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Introduction

- If we hear the word cryptography our first associations might be e-mail encryption, secure website access, smart cards for banking applications or code breaking during world war II, such as the famous attack against the German Enigma encryption machine.

- Cryptography seems closely linked to modern electronic communication.

Introduction

- However, cryptography is a rather old business, with early examples dating back to about 2000 B.C., when non-standard "secret" hieroglyphics were used in ancient Egypt.
- Since Egyptian days cryptography has been used in one form of the other in many if not most, cultures that developed written language.
- For instance, there are documented cases of secret writing in ancient Greece, namely the scytale of Sparta, or the famous Caesar cipher in ancient Rome.

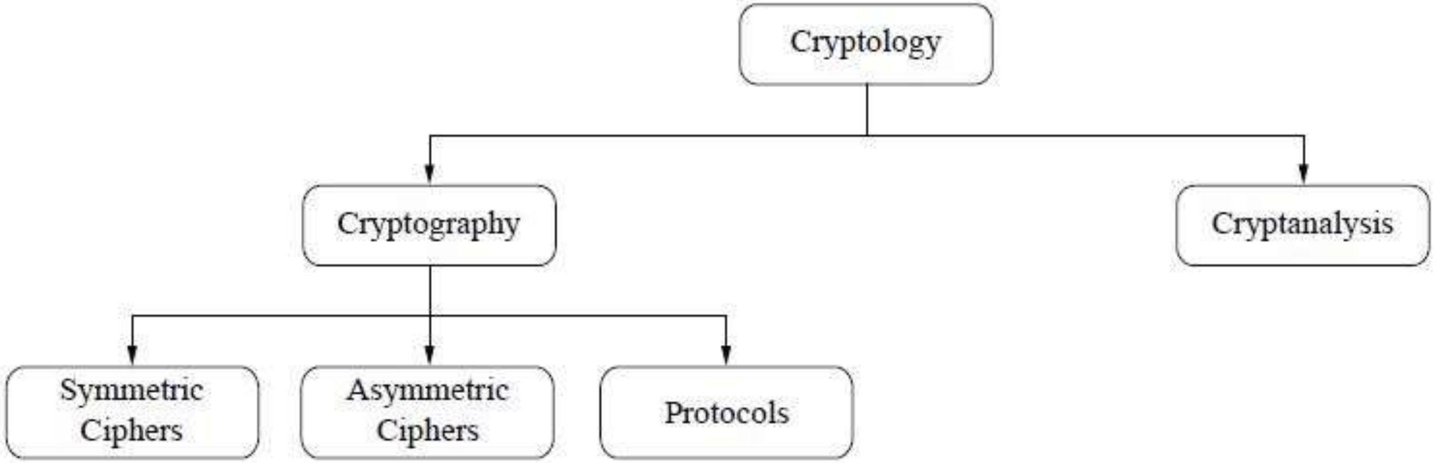
Introduction

- For instance, there are documented cases of secret writing in ancient Greece, namely the scytale of Sparta, or the famous Caesar cipher in ancient Rome.

- This courses, however, strongly focuses on modern cryptographic methods and also teaches many data security issues and their relationship with cryptography.

Cryptography

- Let’s now have a look at the field of cryptography.
- The first thing that we notice is that the most general term is cryptology and not cryptography



Cryptography

- Cryptology splits into two main branches:
  - Cryptography: Science of secret writing
  - Cryptanalysis: The science and sometimes art of breaking cryptosystems

- Cryptanalysis is of central importance for modern cryptosystems: without people wo try to break

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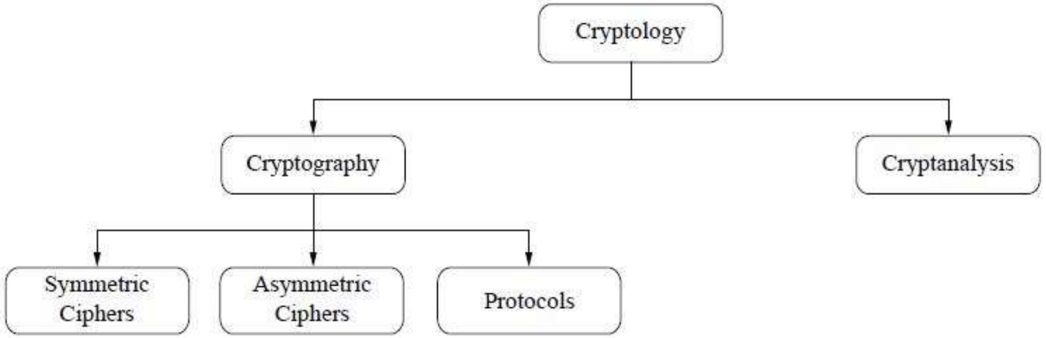

Cryptanalysis

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- The focus in this course is on crypto**graphy**: We introduce most important practical crypto algorithms in detail.

- These are all crypto algorithms that have withstood cryptanalysis for a long time, in most cases for several decades.

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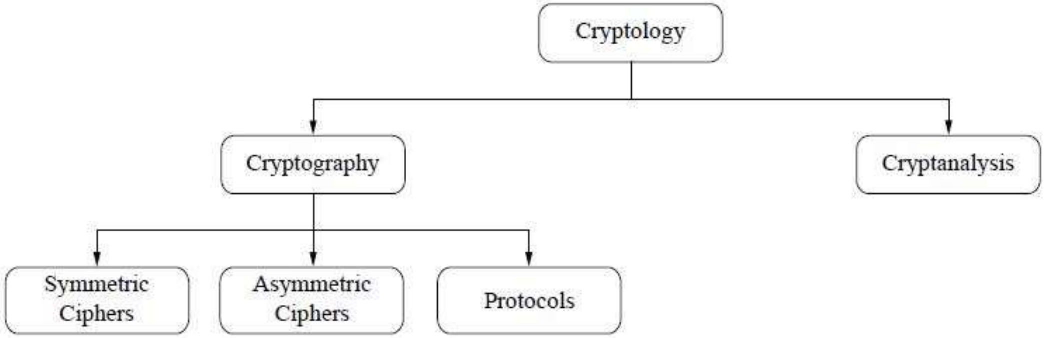
Cryptanalysis

- In the case of crypt**analysis** we will mainly restrict ourselves to providing state-of-the-art results with respect to breaking the crypto algorithms that are introduced, e.g., the factoring record for breaking the RSA scheme.

- Let's now go back to our Figure:



Cryptography

- Cryptography itself splits into three main branches:
  - Symmetric Algorithms
  - Asymmetric (or Public-key) Algorithms
  - Cryptographic protocols

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Symmetric Cryptography

- Symmetric Algorithms are what many people assume cryptography is about: two parties have an encryption and decryption method for which they share a secret key.

- All cryptography from ancient times until 1976 was exclusively based on symmetric methods.

- Symmetric ciphers are still in widespread use, especially for data encryption and integrity check of messages.

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Asymmetric Cryptography

- Asymmetric (or Public-Key) Algorithms In 1976 an entirely different type of cipher was introduced by Whitfield Diffie, Martin Hellman and Ralph Merkle.
- In public-key cryptography, a user possesses a secret key as in symmetric cryptography but also a public key.
- Asymmetric algorithms can be used for applications such as digital signatures and key establishment, and also for classical data encryption.

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Cryptographic Protocols

- Roughly speaking, crypto protocols deal with the application of cryptographic algorithms.
- Symmetric and asymmetric algorithms can be viewed as building blocks with which applications such as secure Internet communication can be realized.
- The Transport Layer Security (TLS) scheme, which is used in every Web browser, is an example of a cryptographic protocol.

Cryptographic Protocols

- Strictly speaking, hash functions, form a third class of algorithms but at the same time they share some properties with symmetric ciphers.

- In the majority of cryptographic applications in practical systems, symmetric and asymmetric algorithms (and often also hash functions) are all used together.

- This is sometimes referred to as *hybrid schemes*.

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Cryptographic Protocols

- The reason for using both families of algorithms is that each has specific strengths and weaknesses.

- The main focus of this book is on symmetric and asymmetric algorithms, as well as hash functions.

Cryptographic Protocols

- However, we will also introduce basic security protocols.
- In particular, we will introduce several key establishment protocols and what can be achieved with crypto protocols: confidentiality of data, integrity of data, authentication of data, user identification, etc.

Symmetric Cryptography

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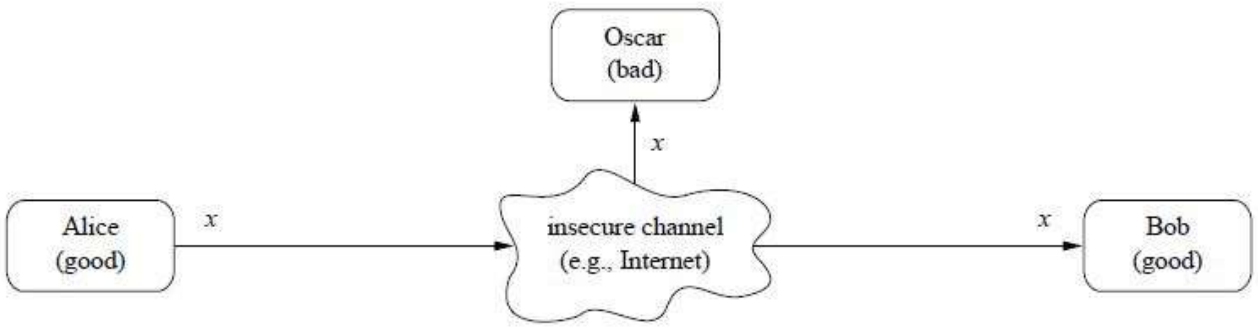
Introduction

- Symmetric cryptographic schemes are also referred to as *symmetric-key*, *secret-key*, and *single-key* schemes or algorithms.
- Symmetric cryptography is best introduced with an easy to understand problem:
- There are two users, Alice and Bob, who want to communicate over

an insecure *channel*.



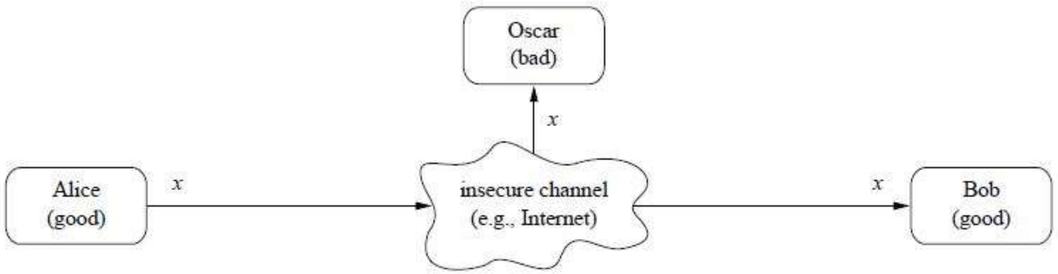
Introduction

- The term channel might sound a bit abstract but it is just a general term for the communication link: This can be the Internet, a stretch of air in the case of mobile phones or wireless LAN communication, or any other communication media you can think of.
- The actual problem starts with the bad guy, Oscar, who has access to the channel, for instance, by hacking into

an Internet router or by

listening to the radio signals

of a Wi-Fi communication.



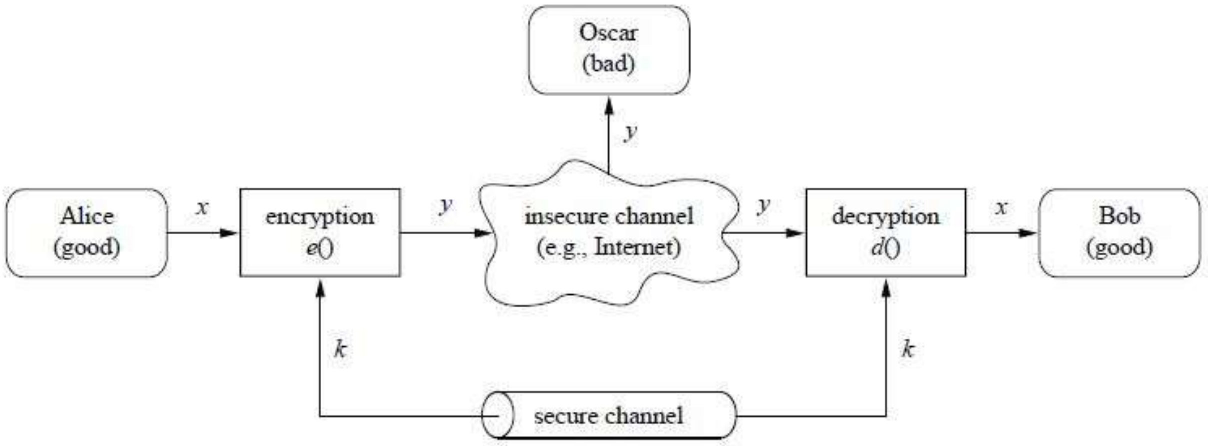
Introduction

- This type of unauthorized listening is called *eavesdropping*.
- Obviously, there are many situations in which Alice and Bob would prefer to communicate without Oscar listening.
- For instance, if Alice and Bob represent two offices of a car manufacturer, and they are transmitting documents containing the business strategy for the introduction of new car models in the next few years, these documents should not get into the hands of their competitors, or of foreign intelligence agencies for that matter.

Basics

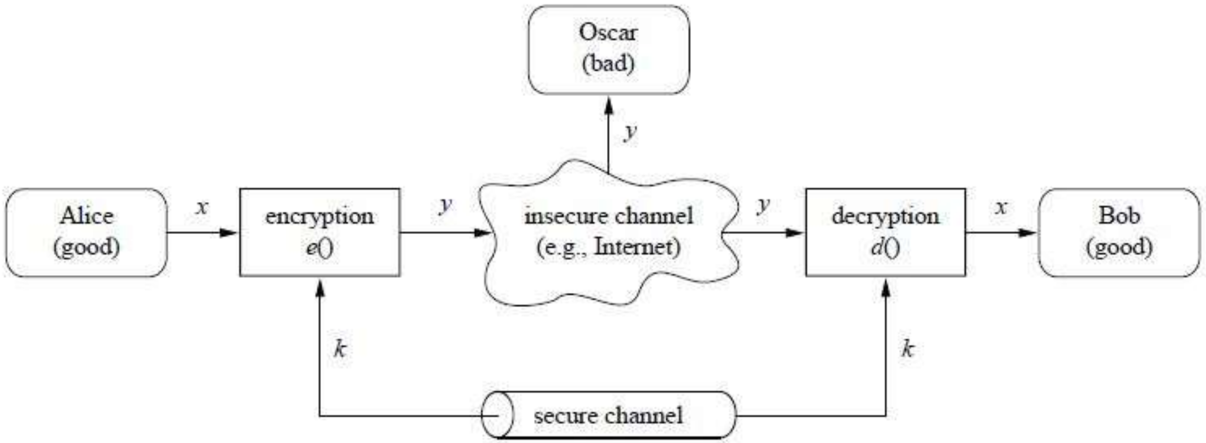
- In this situation, symmetric cryptography offers a powerful solution: Alice encrypts her message *x* using a symmetric algorithm, yielding the ciphertext *y*.
- Bob receives the ciphertext and decrypts the message. Decryption is, thus, the inverse

process of encryption



Basics

- What is the advantage?
- If we have a strong encryption algorithm, the ciphertext will look like random bits to Oscar and will contain no information whatsoever that is useful to him.



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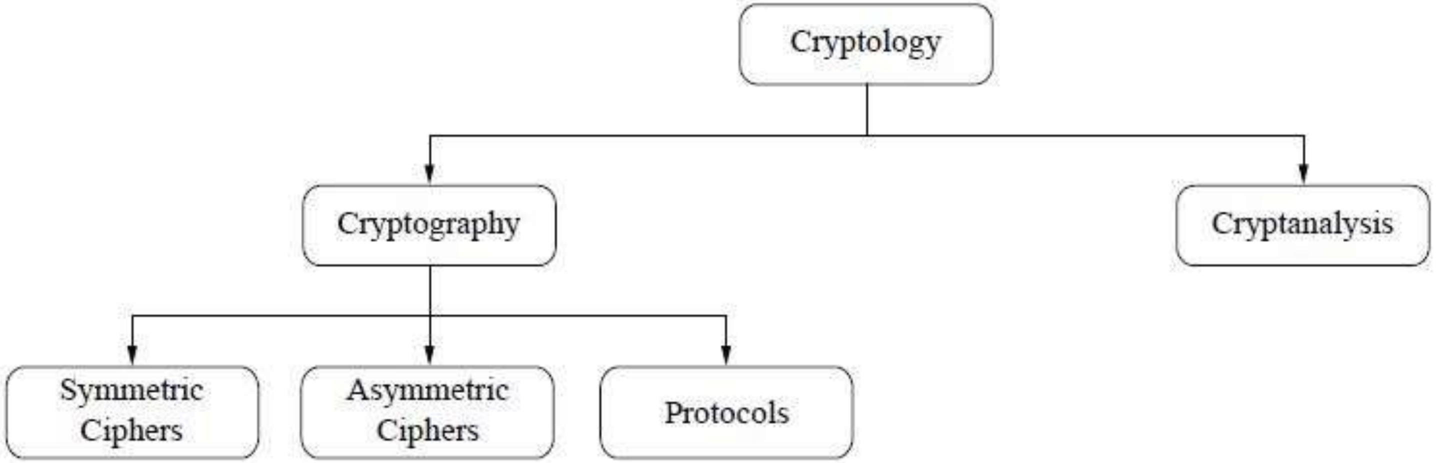

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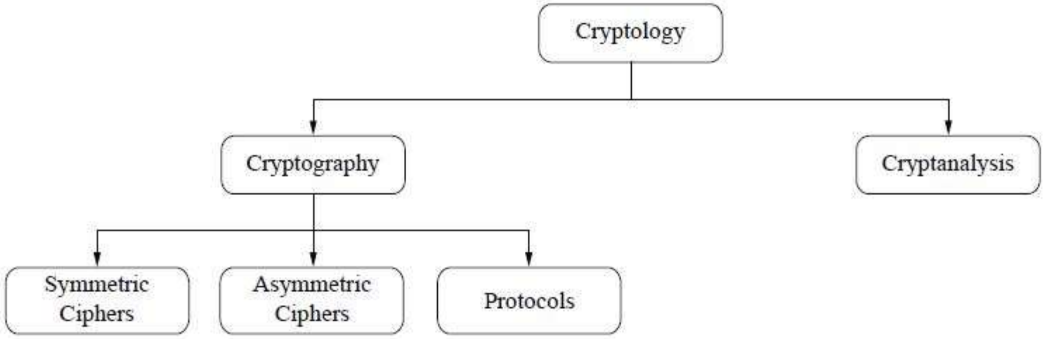

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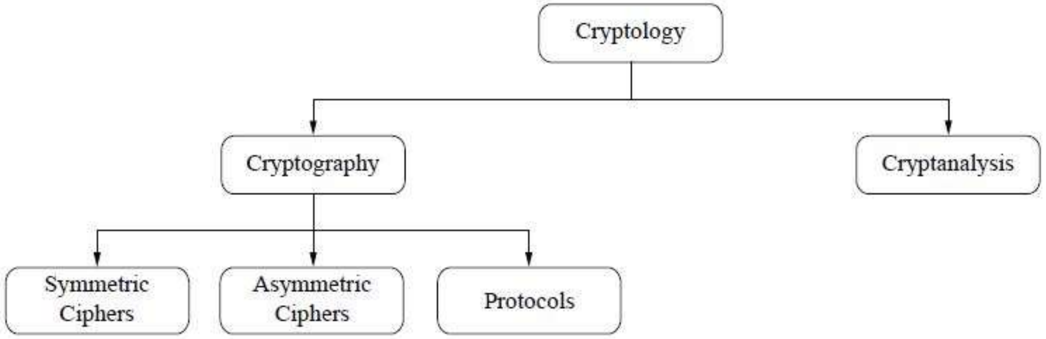

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The Substitution Cipher

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Introduction

- We will now learn one of the simplest methods for encrypting text, the *substitution (= replacement) cipher*.
- Historically this type of cipher has been used many times, and it is a good illustration of basic cryptography.
- We will use the substitution cipher for learning some important facts about key lengths and about different ways of attacking ciphers.

Introduction

- The goal of the substitution cipher is the encryption of text (as opposed to bits in modern digital systems).

- The idea is very simple: We substitute each letter of the alphabet with another one.

Example

- $A \rightarrow k$, $B \rightarrow d$, $C \rightarrow W$ and so on.
- For instance, the pop group ABBA would be encrypted as kddk.
- We assume that we choose the substitution table completely randomly, so that an attacker is not able to guess it.
- Note that the substitution table is the key of this cryptosystem

Notes

- Note that the substitution table is the key of this cryptosystem.
- As always in symmetric cryptography, the key has to be distributed between Alice and Bob in a secure fashion.

Example

- Let's look at another ciphertext:
  - iq ifcc vqqr fb rdq vfllcq na rdq cfjwhwz hr bnnb hcc hwwhbsqvqbre hwq vhlq
- This does not seem to make too much sense and looks like decent cryptography.
- However, the substitution cipher is not secure at all. Let's look at ways of breaking the cipher.

First Attack: Brute-Force

- *Brute-force* attacks are based on a simple concept: Oscar, the attacker, has the ciphertext from eavesdropping on the channel and happens to have a short piece of plaintext, e.g., the header of a file that was encrypted.
- Oscar now simply decrypts the first piece of ciphertext with *all possible* keys.
- Again, the key for this cipher is the substitution table.
- If the resulting plaintext matches the short piece of plaintext, he knows that he has found the correct key.

Definition

- Let (x,y) denote the pair of plaintext and ciphertext and let K ={k_1, k_2, ..., k_k} be the key space of all possible keys k_i. A brute-force attack now checks for every k_i ∈ K if $d_{k_i}(y) = x$ holds,

- If the equality holds a possible correct key is found: if not, proceed with the next key.

Brute-Force

- In practice, a brute-force attack can be more complicated because incorrect keys can give false positive results.
- It is important to note that a brute-force attack against symmetric ciphers is always possible *in principle*.
- Whether it is feasible in practice depends on the key space, i.e., on the number of possible keys that exist for a given cipher.
- If testing all the keys on many modern computers takes too much time, i.e., several decades, the cipher is *computationally secure* against a brute-force attack.

Brute-Force

- Let's determine the key space of the substitution cipher: When choosing the replacement for the first letter A, we randomly choose one letter from the 26 letters of the alphabet (in the example above we chose k).
- The replacement for the next alphabet letter B was randomly chosen from the remaining 25 letters, etc.
- Thus there exist the following number of different substitution tables:

Brute-Force

- Thus there exist the following number of different substitution tables:
- Key space of the substitution cipher = $26 \cdot 25 \cdot 24 \cdots 2 \cdot 1 = 26!$
- This is approximately 2^{88}!

Brute-Force

- Even with hundreds of thousands of high-end PCs such a search would take several decades!
- Thus, we are tempted to conclude that the substitution cipher is secure.
- But this is incorrect because there is another, more powerful attack.

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Letter Frequency Analysis

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Introduction

- First we note that the brute-force attack from above treats the cipher as a black box, i.e., we do not analyse the internal structure of the cipher.

- The substitution cipher can easily be broken by such an analytical attack.

Introduction

- That mans that the statistical properties of the plaintext are preserved in the ciphertext
- If we go back to the following example: iq ifcc vqqr fb rdq vfllcq na rdq cfjwhwz hr bnnb hcc hwwhbsqvqbre hwq vhlq we observe that the letter q occurs most frequently in the text.
- From this we know that q must be the substitution for one of the frequent letters in the English language.

Letter Frequency Analysis

- For practical attacks, the following properties of language can be exploited:
- 1) Determine the frequency of every ciphertext letter.
- The frequency distribution, often even of relatively short pieces of encrypted text, will be close to that of the given language in general.
- In particular, the most frequent letters can often easily be spotted in ciphertexts.
- For instance, in English E is the most frequent letter (about 13%),
- T is the second most frequent letter (about 9%),
- A is the third most frequent letter (about 8%), and so on.
- Table 1 lists the letter frequency distribution of English

Table 1.1

Letter	Frequency	Letter	Frequency
A	0.0817	N	0.0675
B	0.0150	O	0.0751
C	0.0278	P	0.0193
D	0.0425	Q	0.0010
E	0.1270	R	0.0599
F	0.0223	S	0.0633
G	0.0202	T	0.0906
H	0.0609	U	0.0276
I	0.0697	V	0.0098
J	0.0015	W	0.0236
K	0.0077	X	0.0015
L	0.0403	Y	0.0197
M	0.0241	Z	0.0007

Letter Frequency Analysis

- 2) The method above can be generalised by looking at pairs or triples or quadruples, and so on of ciphertext symbol.s

- For instance, in English (and some other European languages), the letter Q is almost always followed by a U.

- This behaviour can be exploited to detec the substitution of the letter Q and the letter U.

Letter Frequency Analysis

- 3) If we assume that word separators (blanks) have been found (which is only sometimes the case), one can often detect frequent short words such as THE, AND, etc.
- Once we have identified one of these words, we immediately know three letters (or whatever the length of the word is) for the entire text.

Example

- In practice, the three techniques listed above are oftedn combined to break substitution ciphers
- If we analyse the encrypted text:
  - iq ifcc vqqr fb rdq vfllcq na rdq cfjwhwz hr bnnb hcc hwwhbsqvqbre hwq vhlq
- We obtain: WE WILL MEET IN THE MIDDLE OF THE LIBRARY AT NOON ALL ARRANGMENTS ARE MADE

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Cryptanalysis

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Introduction

- This class looks at recommended key lengths of symmetric ciphers and different ways of attacking crypto algorithms

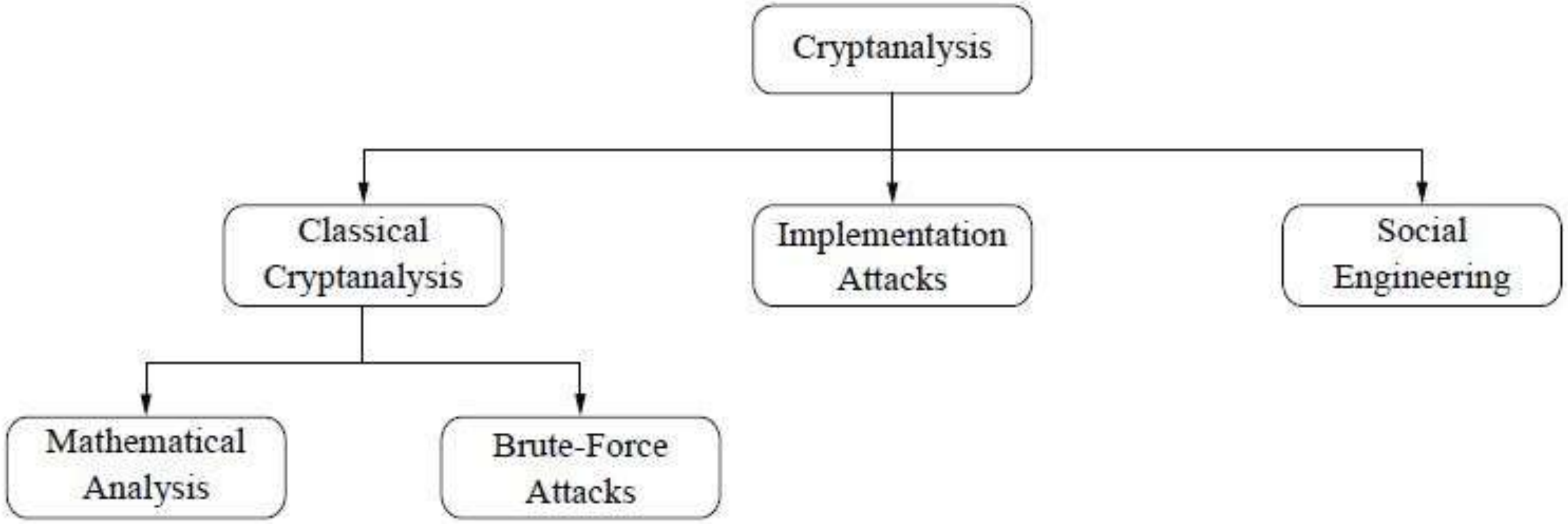
- It is stressed that a cipher should be secure even if the attacker knows the details of the algorithm

General Thoughts on Breaking Cryptosystems

- If we ask someone with some technical background what breaking ciphers is about, he/she will most likely say that code breaking has to do with heavy mathematics, smart people and large computers.
- We have images in mind of the British code breakers during World War II, attacking the German Enigma cipher with extremely smart mathematicians (the famous computer scientist Alan Turing headed the efforts) and room-sized electro-mechanical computers.
- However, in practice there are also other methods of code breaking.

Letter Frequency Analysis

- Let's look at different ways of breaking cryptosystems in the real world



Classical Cryptanalysis

- Classical cryptanalysis is understood as the science of recovering the plaintext x from the ciphertext y, or, alternatively, recovering the key k from the ciphertext y.
- We recall from the earlier discussion that cryptanalysis can be divided into:
  - analytical attack: which exploit the internal structure of the encryption method,
  - brute-force attacks: which treat the encryption algorithm as a black box and test all possible keys.

Implementation Attacks

- Side-channel analysis can be used to obtain a secret key, for instance, by measuring the electrical power consumption of a processor which operates on the secret key.
- The power trace can then be used to recover the key by applying signal processing techniques.
- In addition to power consumption, electromagnetic radiation or the runtime behaviour of algorithms can give information about the secret key and are, thus, useful side channels.
- Note also that implementation attacks are mostly relevant against cryptosystems to which an attacker has physical access, such as smart cards.
- In most Internet-based attacks against remote systems, implementation attacks are usually not a concern.

Social Engineering Attacks

- Bribing, blackmailing, tricking or classical espionage can be used to obtain a secret key by involving humans.
- For instance, forcing someone to reveal his/her secret key, e.g., by holding a gun to his/her head can be quite successful.
- Another, less violent, attack is to call people whom we want to attack on the phone, and say: "This is the IT department of your company. For important software updates we need your password".
- It is always surprising how many people are naive enough to actually give out their passwords in such situations.

Notes

- This list of attacks against cryptographic system is certainly not exhaustive.
- For instance, buffer overflow attacks or malware can also reveal secret keys in software systems.
- You might think that many of these attacks, especially social engineering and implementation attacks, are “unfair,” but there is little fairness in real-world cryptography.
- If people want to break your IT system, they are already breaking the rules and are, thus, unfair.

Notes

- The major point to learn here is: An attacker always looks for the weakest link in your cryptosystem.
- That means we have to choose strong algorithms *and* we have to make sure that social engineering and implementation attacks are not practical.
- Even though both implement attacks and social engineering attacks can be quite powerful in practice, in this course we mainly focus on attacks based on mathematical cryptanalysis.

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Kerckhoff’s Principle

- Solid cryptosystems should adhere to Kerckhoff’s Principle, postulated by Auguste Kerckhoff's in 1883:

- A cryptosystem should be secure even if the attacker (Oscar) knows all details about the system, with the exception of the secret key. In particular, the system should be secure when the attacker knows the encryption and decryption algorithms.

Kerckhoff’s Principle Notes

- Kerckhoffs’ Principle is counterintuitive! It is extremely tempting to design a system which appears to be more secure because we keep the details hidden.
- This is called *security by obscurity*.
- However, experience and military history has shown time and again that such systems are almost always weak, and they are very often broken easily as soon as the secret design has been reverse-engineered or leaked out through other means.

Kerckhoff’s Principle Notes

- An example is the Content Scrambling System (CSS) for DVD content protection, which was broken easily once it was reverse engineered.
- This is why a cryptographic scheme must remain secure even if its description becomes available to an attacker.

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How many Key bits are enough?

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Introduction

- During the 1990s there was much public discussion about the key length of ciphers.
- Before we provide some guidelines, there are two crucial aspects to remember:
- 1. The discussion of key lengths for symmetric crypto algorithms is only relevant if a brute-force attack is the best-known attack.
- As we saw in previous classes during the security analysis of the substitution cipher, if there is an analytical attack that works, a large key space does not help at all.
- Of course, if there is the possibility of social engineering or implementation attacks, a long key also does not help.

Introduction

- The key lengths for symmetric and asymmetric algorithms are dramatically different.
- For instance, an 80-bit symmetric key provides roughly the same security as a 1024-bit RSA (RSA is a popular asymmetric algorithm) key.
- Both facts are often misunderstood, especially in the semitechnical literature.
- Table 2 gives a rough indication of the security of symmetric ciphers *with respect to brute-force attacks*.

Table

- As described in previous class, a large key space is a necessary but not sufficient condition for a secure symmetric cipher.
- The cipher must also be strong against analytical attacks.

Key length	Security estimation
56–64 bits	short term: a few hours or days
112–128 bits	long term: several decades in the absence of quantum computers
256 bits	long term: several decades, even with quantum computers that run the currently known quantum computing algorithms

Foretelling the future

- Of course, predicting the future tends to be tricky.
- We can't really foresee new technical or theoretical developments with certainty.
- As you can imagine, it is very hard to know what kind of computers will be available in the year 2030.
- For medium-term predictions, Moore's Law is often assumed.
- Roughly speaking, Moore's law states that computing power doubles every 18 months while the cost stays constant.

Foretelling the future

- This has the following implications in cryptography:
- If today we need one moth and computers work $1,000,000 to break a cipher X, then:
  - The cost for breaking the cipher will be $500,000 in 18 months
  - $250,000 in 3 years
  - $125,000 in 4.5 years, and so on.
- We only have to buy half as many computers each time.

Foretelling the future

- It is important to stress that Moore's Law is an exponential function.
- In 15 years, i.e., after 10 iterations of computer power doubling, we can do $2^{10} = 1024$ as many computations for the same money as we would need to spend today.
- Stated differently, we only need to spend about 1/1000th of today's money to do the same computation.
- In the example above that means that we can break cipher X in 15 years within one month at a cost of about \$1,000,000/1024≈\$1000.

Foretelling the future

- Alternatively, with $1,000,000, an attack can be accomplished within 45 minutes in 15 years from now.
- Moore's Law behaves similarly to a bank account with a 50% interest rate: The compound interest grows very, very quickly.
- Unfortunately, there are few trustworthy banks which offer such an interest rate.

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Divisibility

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Introduction

- We are going to look at two historical ciphers so that we can understand modular arithmetic with integers.
- Even though the historical ciphers are no longer relevant, modular arithmetic is extremely important in modern cryptography, especially for asymmetric algorithms.
- Ancient ciphers date back to Egypt, where substitution ciphers were used.
- A very popular special case of the substitution cipher is the Caesar cipher, which is said to have been used by Julius Caesar to communicate with his army.

Introduction

- The Caesar cipher simply shifts the letters in the alphabet by a constant number of steps.
- When the end of the alphabet is reached, the letters repeat in a cyclic way, similar to numbers in modular arithmetic.
- To make computations with letters more practicable, we can assign each letter of the alphabet a number.

Introduction

- By doing so, an encryption with the Caesar cipher simply becomes a (modular) addition with a fixed value. Instead of just adding constants, a multiplication with a constant can be applied as well.
- This leads us to the *affine cipher*.
- Both the Caesar cipher and the affine cipher will now be discussed in more detail.

Definition

- Let $a, b \in \mathbb{Z}$ then a divides b, denoted $a|b$ if there exists $c \in \mathbb{Z}$ such that b=ac.
- If $a|b$ then a is said to be a divisor or factor of b.
- The notation $a \nmid b$ means that a does not divide b.

Example

- $3|6$ since there exists $c \in \mathbb{Z}$ such that 6=3c. Here, c=2. Hence 3 is a divisor of 6.
- $3 \nmid 5$ since there does not exist $c \in \mathbb{Z}$ such that 5=3c.
- Note that c would be $\frac{5}{3}$ here and $\frac{5}{3} \notin \mathbb{Z}$. Hence 3 is not a divisor of 5.
- $3|-6$ since there exists $c \in \mathbb{Z}$ such that -6=3c. Here, c=-2. Hence 3 is a divisor of -6.

Example

- If a ϵ $\mathbb{Z}$, then $a|0$ since there exists c ϵ $\mathbb{Z}$ such that 0=ac. Here, c=0.
- In other words, any integer divides 0.
- If a ϵ $\mathbb{Z}$ and $0|a$, then there exists c ϵ $\mathbb{Z}$ such that a=0c if and only if a=0.
- In other words, the only integer having zero as a divisor is zero.

Theorem D1

- Let $a, b, c \in \mathbb{Z}$, if $a|b, b|c$ then $a|c$
- Proof:

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Theorem D2

- Let $a, b, c, m, n \in \mathbb{Z}$, if c|a, $c|b$ then $c|ma + nb$
- Proof:
- The expression ma+nb is said to be an integral linear combination of and an b.

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Theorem D3

- The division algorithm: Let $a, b \in \mathbb{Z}$ with b>0. Then there exists unique $q, r \in \mathbb{Z}$ such that $a = bq + r$ where $0 \leq r < b$

- Note: q stands for quotient and r for remainder

Notes

- This algorithm is saying that any integer can be represented by another integer times it’s quotient plus a remainder.

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Prime Numbers

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Introduction

- Prime numbers play a key role in cryptography.
- Every integer greater than one has at least two positive divisors, namely 1 and the integer itself.
- Those positive integers having no other positive divisors are of crucial importance.

Definition

- Let $p \epsilon \mathbb{Z}$ with p>1. Then p is said to be a prime if the only positive divisors of p are 1 and p.
- If $n \in \mathbb{Z}$, n>1, and n is not prime, then n is said to be composite.
- Note that the positive integer 1 is neither prime nor composite by definition.

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Example

- The prime numbers between 2 and 50 inclusive are:
- 2,3,5,7,11,13,17,19,23,29,31,37,43 and 47.
- This example shows that all prime numbers except 2 are odd and hence the integer 2 is the only prime even number.

Notes

- Having introduced the concept of a prime number, a most fundamental question arises.

- Are there finitely many or infinitely many prime numbers?

Theorem P2-P4

- Theorem P2: Every integer greater than 1 has a prime divisor.

- Theorem P3: There are infinitely many prime numbers.

- Theorem P4: Let n be a composite number. Then n has a prime divisor p with $p \leq \sqrt{n}$.

Example

- Suppose that we wish to find all prime numbers less than or equal to 50.
- By Theorem P4, any composite number less than or equal to 50 must have a prime divisor less than or equal to $\sqrt{50} \approx 7.07$.
- The prime number less than or equal to 7.07 are 2,3,5 and 7.
- Hence, form a list of the integers from 2 to 5 and delete all multiples of 2, 3, 5 and 7.

Example

- Note that all such multiples are clearly composite.

	2	3	~~4~~	5	~~6~~	7	~~8~~	~~9~~	~~10~~
11	~~12~~	13	~~14~~	~~15~~	~~16~~	17	~~18~~	19	~~20~~
~~21~~	~~22~~	23	~~24~~	~~25~~	~~26~~	~~27~~	~~28~~	29	~~30~~
31	~~32~~	~~33~~	~~34~~	~~35~~	~~36~~	37	~~38~~	~~39~~	~~40~~
41	~~42~~	43	~~44~~	~~45~~	~~46~~	47	~~48~~	~~49~~	~~50~~

- Any number remaining in the list is not divisible by 2,3,5 and 7 are therefore cannot be composite.
- Thus the numbers remaining in the list are prime.

Algorithm

- In addition to finding prime numbers, Theorem P4 can be used to determine an algorithm for testing whether a given positive integer n>1 is prime or composite.
- One check whether n is divisible by any prime number p with $p \leq \sqrt{n}$.
- If it is divisible, then n is composite; if not, then n is prime.
- Such an algorithm is an example of primality test.

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